

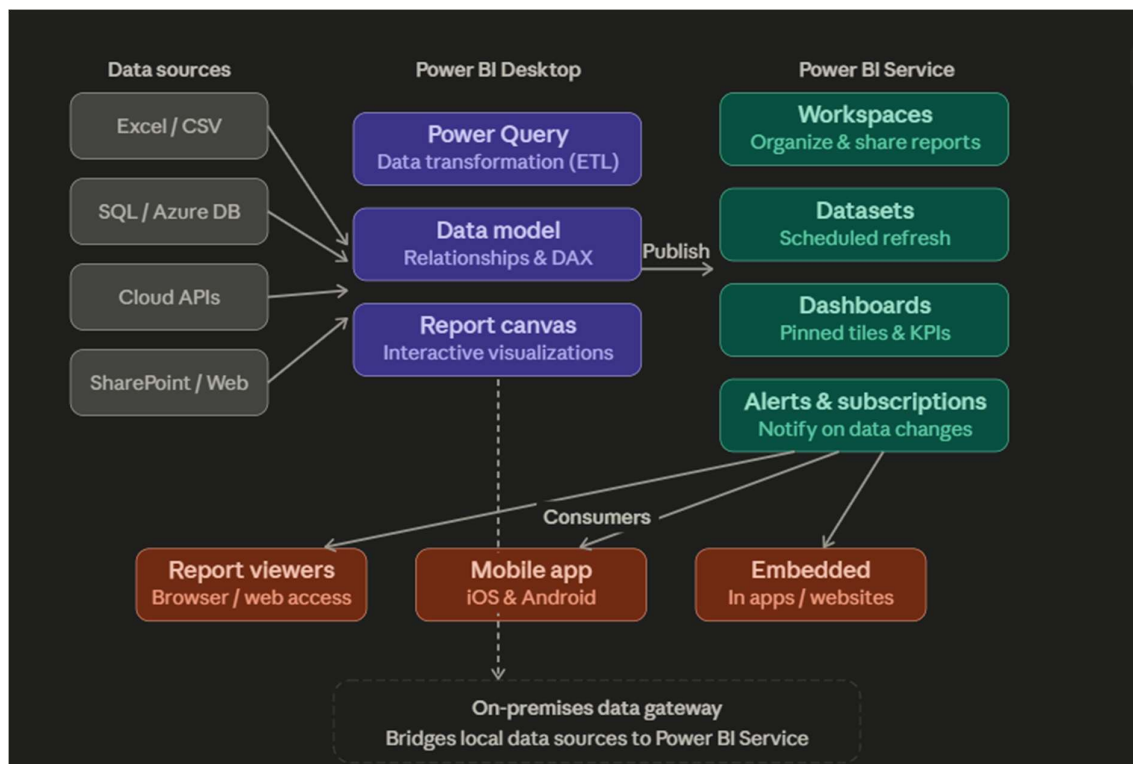
Experiment-5

AIM: Study Power BI and its functionalities.

Power BI:

Power BI is Microsoft's business intelligence and data visualization platform. It allows users to connect to hundreds of data sources, transform raw data, build interactive dashboards, and share insights across an organization — all without needing to write code (though code-based options exist for advanced users).

It comes in three main forms: **Power BI Desktop** (a Windows app for building reports), **Power BI Service** (a cloud platform for sharing and collaboration), and **Power BI Mobile** (for viewing reports on smartphones and tablets).



Core Functionalities:

1. Data Connectivity

Power BI connects to 100+ data sources — from flat files (Excel, CSV), relational databases (SQL Server, MySQL, PostgreSQL), cloud services (Azure, Salesforce, Google Analytics), and even web pages and APIs. You can load data in **import mode** (snapshot) or **DirectQuery mode** (live query against the source).

2. Power Query — ETL Layer

The built-in ETL (Extract, Transform, Load) engine. You use it to clean, reshape, and combine data before it reaches your model. Operations include filtering rows, splitting columns, merging tables, unpivoting data, and writing custom **M language** transformations. Everything you do is recorded as a reproducible set of steps.

3. Data Modeling

Power BI uses an in-memory columnar engine (**VertiPaq**). You define relationships between tables (one-to-many, many-to-many), set cardinality, and build a star or snowflake schema. The model is the foundation all visuals query against.

4. DAX — Data Analysis Expressions

DAX is the formula language used to create **calculated columns** and **measures** (aggregated calculations). It resembles Excel formulas but is designed for relational data. Common functions include CALCULATE, SUMX, FILTER, RELATED, DATEADD, and time-intelligence functions like TOTALYTD.

5. Visualizations

Power BI has 30+ built-in chart types — bar, line, pie, scatter, map, treemap, gauge, KPI cards, matrix tables, and more. You can also import **custom visuals**

from AppSource (marketplace) or build your own with **R** or **Python** scripts. Visuals are interactive by default — clicking one filters others on the same page.

6. Reports vs. Dashboards

A **report** is a multi-page document created in Power BI Desktop with rich analytical detail. A **dashboard** is a single-page canvas in Power BI Service made of "pinned tiles" from multiple reports — designed for executive-level at-a-glance monitoring with real-time data alerts.

7. Row-Level Security (RLS)

You can restrict which rows of data a user sees based on their identity. For example, a sales manager sees only their region's data. RLS rules are defined in Power BI Desktop and enforced in the Service.

8. Sharing & Collaboration

Reports are published to Power BI Service, organized in **workspaces**, and shared via apps, direct links, or embedded in Microsoft Teams, SharePoint, or external websites (Power BI Embedded).

Key Terms:

- **Workspace** — A folder-like container in the Service for organizing reports and datasets
- **Dataset** — The published data model that reports connect to
- **Gateway** — Software that lets cloud-based Power BI refresh on-premises data
- **Dataflow** — Reusable Power Query transformations stored in the cloud

- **Paginated Reports** — Pixel-perfect, printable reports (like invoices) built with Power BI Report Builder
-

Practical Workflow (End to End)

1. Connect to your data source in **Power BI Desktop**
 2. Clean and shape data using **Power Query Editor**
 3. Build relationships in the **Model view**
 4. Write measures using **DAX**
 5. Design report pages on the **Report canvas**
 6. **Publish** to Power BI Service
 7. Pin visuals to a **Dashboard**
 8. Set up **scheduled refresh** and share with stakeholders
-

Experiment-6

AIM: Perform data transformation Operation on Table in Power Query editor.

Operations Performed:

1. Removed unnecessary column:

Removed column named “id” – not useful for analysis

2. Renamed Columns:

- price → **House Price**
- sqft_living → **Living Area**
- sqft_lot → **Lot Area**
- yr_built → **Year Built**

3. Changed data types:

- price → Decimal Number
- bedrooms → Whole Number
- bathrooms → Decimal
- zipcode → Text

4. Replaced Values:

Column: waterfront

Values:

- 0 → No
- 1 → Yes

5. Transformed date columns:

From 20141013T000000 to 13-10-2014

6. Removed Duplicates:

From columns – zipcode and price

Final Transformed Dataset:

Table.Distinct(#Renamed Columns1, {"yr_built"})

	Date	House Price	Bedrooms	Bathrooms	Living Area	Lot Area	Floors
1	13-10-2014	231300	2	1.00	1180	5650	
2	09-12-2014	538000	3	2.25	2570	7242	
3	25-02-2015	180000	2	1.00	770	10000	
4	09-12-2014	604000	4	3.00	1960	5000	
5	18-02-2015	510000	3	2.00	1680	8080	
6	12-05-2014	1230000	4	4.50	5420	101930	
7	27-06-2014	257500	3	2.25	1715	6819	
8	15-01-2015	291850	3	1.50	1060	9711	
9	15-04-2015	229500	3	1.00	1780	7470	
10	12-03-2015	323000	3	2.50	1890	6560	
11	27-05-2014	468000	2	1.00	1160	6000	
12	12-03-2015	530000	5	2.00	1810	4850	
13	24-01-2015	650000	4	3.00	2950	5000	
14	31-07-2014	395000	3	2.00	1890	14040	
15	29-05-2014	485000	4	1.00	1600	4300	
16	05-12-2014	128900	2	1.00	1200	9850	
17	14-05-2014	385000	4	1.75	1630	6980	
18	26-08-2014	200000	3	2.75	3050	44867	
19	16-05-2014	252700	2	1.50	1070	9643	
20	26-06-2014	937000	3	1.75	2450	2691	
21	01-12-2014	667000	3	1.00	1400	1581	
22	02-03-2015	719000	4	2.50	2570	7175	
23	24-06-2014	687500	4	1.75	2330	5000	
24	03-12-2014	322500	4	2.75	2060	6659	
25	13-02-2015	240000	4	1.00	1220	8075	
26	20-06-2014	605000	4	2.50	2620	7553	
27	11-08-2014	775000	4	2.25	4220	24186	

20 COLUMNS, 53 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:50 PM

Experiment-7

AIM: Create Table Using DAX.

DAX Code Used:

Summary_Table =

```
SUMMARIZE(  
    'Date Table',  
    'Date Table'[Month Short Name],  
    'Date Table'[WeekDay short Name],  
    "Total Value", SUM('Date Table'[Month Number])  
)
```

The screenshot shows the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, displaying the DAX formula bar with the following code:

```
1 Summary_Table =  
2 SUMMARIZE(  
3     'Date Table',  
4     'Date Table'[Month Short Name],  
5     'Date Table'[WeekDay short Name],  
6     "Total Value", SUM('Date Table'[Month Number])  
7 )
```

Below the formula bar, a preview of the 'Summary_Table' is shown. It has three columns: 'Month Short Name', 'WeekDay short Name', and 'Total Value'. The table contains 84 rows of data, showing the total value for each month and day combination. A tooltip is visible over the 'Summary_Table' in the 'Data' pane, showing its name, storage mode (Import), and the last refresh time (2/5/2026, 9:50:22 pm).

Month Short Name	WeekDay short Name	Total Value
Jan	Sat	5
Feb	Sat	8
Mar	Sat	12
Apr	Sat	20
May	Sat	20
Jun	Sat	24
Jul	Sat	35
Aug	Sat	32
Sep	Sat	36
Oct	Sat	50
Nov	Sat	44
Dec	Sat	60
Jan	Sun	6
Feb	Sun	8
Mar	Sun	12
Apr	Sun	16
May	Sun	25
Jun	Sun	24
Jul	Sun	35
Aug	Sun	32
Sep	Sun	36
Oct	Sun	50
Nov	Sun	44

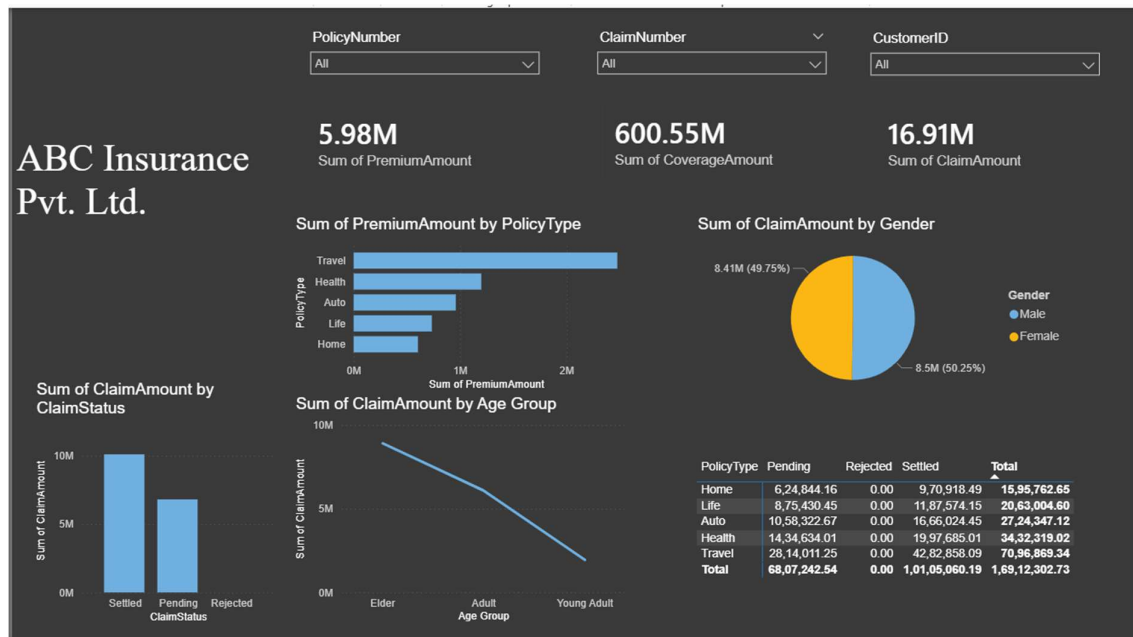
Experiment - 8

AIM: Create dashboard in PowerBI.

Raw Data:

PolicyNum	CustomerID	Gender	Age	PolicyType	PolicyStart	PolicyEnd	PremiumA	CoverageA	ClaimNum	ClaimDate	ClaimAmo	ClaimStatus
P1	C1	Female	73	Auto	#####	#####	240.64	33175.1	C1	NULL	0	Rejected
P2	C2	Male	44	Travel	#####	#####	1059.73	85046.42	C2	#####	1493.06	Pending
P3	C3	Female	28	Travel	#####	#####	1019.59	68525.53	C3	NULL	0	Rejected
P4	C4	Male	85	Travel	#####	#####	549.7	17053.27	C4	#####	5021.34	Pending
P5	C5	Female	57	Travel	#####	#####	841.15	55007.27	C5	#####	1347.46	Pending
P6	C6	Male	59	Travel	#####	#####	266.65	20899.24	C6	#####	2986.86	Settled
P7	C7	Female	36	Travel	#####	#####	573.25	91251.7	C7	#####	5431.79	Pending
P8	C8	Male	26	Health	#####	#####	925.25	46080.19	C8	#####	2275.24	Settled
P9	C9	Female	42	Travel	#####	#####	1006.22	37869.48	C9	#####	2682.63	Settled
P10	C10	Male	44	Travel	#####	#####	607.42	88203.38	C10	#####	5118.08	Pending
P11	C11	Female	65	Health	#####	#####	333.2	46868.19	C11	NULL	0	Rejected
P12	C12	Male	47	Home	#####	#####	645.73	67012.84	C12	NULL	0	Rejected
P13	C13	Female	46	Travel	#####	#####	646.14	87450.81	C13	NULL	0	Rejected
P14	C14	Male	48	Health	#####	#####	142.27	96524.13	C14	#####	3560.56	Settled
P15	C15	Female	23	Home	#####	#####	398.43	97215.02	C15	NULL	0	Rejected
P16	C16	Male	48	Travel	#####	#####	638.79	49649.72	C16	#####	2081.58	Settled
P17	C17	Female	59	Travel	#####	#####	231.94	92185.53	C17	NULL	0	Rejected
P18	C18	Male	74	Life	#####	#####	302.08	31710.35	C18	NULL	0	Rejected
P19	C19	Female	25	Life	#####	#####	854.05	80203.5	C19	#####	2293.17	Settled
P20	C20	Male	63	Life	#####	#####	440.33	18320.59	C20	#####	4076.85	Pending
P21	C21	Female	57	Health	#####	#####	673.19	35518.09	C21	NULL	0	Rejected
P22	C22	Male	61	Travel	#####	#####	705.9	30736.05	C22	#####	3065.87	Settled
P23	C23	Female	33	Travel	#####	#####	944.45	81644.41	C23	#####	4730.07	Settled
P24	C24	Male	84	Travel	#####	#####	770.64	76113.89	C24	NULL	0	Rejected
P25	C25	Female	53	Home	#####	#####	940.01	80065.78	C25	NULL	0	Rejected

Dashboard created with the given data:



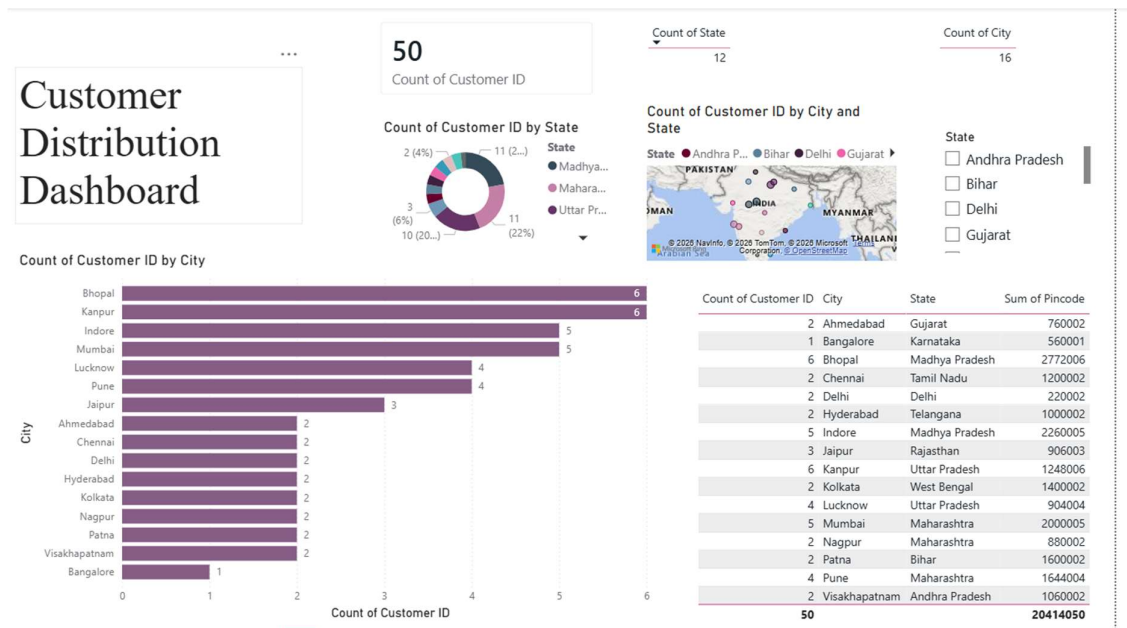
Experiment-9

AIM: Create a Dashboard.

Raw Data:

Customer ID											
Customer	Customer	City	State	Pincode	EmailID	Phone Number					
1	Aarav Singl	Nagpur	Maharash	440001	1Aarav@gr	8.6E+08					
2	Aditi Patel	Bhopal	Madhya Pr	462001	2Aditi@gm	8.97E+08					
3	Aman Verr	Ahmedaba	Gujarat	380001	3Aman@gr	8.34E+08					
4	Ananya Me	Bhopal	Madhya Pr	462001	4Ananya@	8.04E+08					
5	Aniket Sha	Indore	Madhya Pr	452001	5Aniket@g	9.86E+08					
6	Anjali Rao	Bhopal	Madhya Pr	462001	6Anjali@g	9.63E+08					
7	Arjun Joshi	Kanpur	Uttar Prad	208001	7Arjun@gr	9.88E+08					
8	Aryan Gup	Mumbai	Maharash	400001	8Aryan@gr	9.54E+08					
9	Avni Iyer	Kanpur	Uttar Prad	208001	9Avni@gm	8.64E+08					
10	Dhruv Des	Pune	Maharash	411001	10Dhruv@	9E+08					
11	Diya Kapoc	Mumbai	Maharash	400001	11Diya@gr	8.46E+08					
12	Gaurav Na	Pune	Maharash	411001	12Gaurav@	8.82E+08					
13	Isha Reddy	Bhopal	Madhya Pr	462001	13Isha@gr	9.36E+08					
14	Ishaan Sha	Pune	Maharash	411001	14Ishaan@	9.2E+08					
15	Jaya Kuma	Chennai	Tamil Nad	600001	15Jaya@gr	9.24E+08					
16	Kabir Chou	Jaipur	Rajasthan	302001	16Kabir@g	9.73E+08					
17	Kajal Bhatt	Delhi	Delhi	110001	17Kajal@g	8.74E+08					
18	Karan Seth	Visakhapa	Andhra Pr	530001	18Karan@	8.32E+08					
19	Kavya Mish	Indore	Madhya Pr	452001	19Kavya@i	9.23E+08					
20	Krishnan M	Chennai	Tamil Nad	600001	20Krishnar	9.83E+08					
21	Lavanya Ja	Indore	Madhya Pr	452001	21Lavanya	9.09E+08					
22	Manish Du	Bangalore	Karnataka	560001	22Manish@	9.53E+08					
23	Meera Ban	Hyderabad	Telangana	500001	23Meera@	9.2E+08					
24	Mohit Agga	Patna	Bihar	800001	24Mohit@i	8.2E+08					
25	Naina Cha	Kanpur	Uttar Prad	208001	25Naina@	9.63E+08					

Dashboard created:



Experiment - 10

AIM: Create A dashboard.

Raw Data:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
TransactionId	Transaction	Amount	BankName	BankName	Remaining	City	Gender	Transaction	Status	Transaction	DeviceType	PaymentM	Merchant	Purpose	Customer/	PaymentM	Currency	Customer/	Merchant	AccountNumber	
1	TXN00001	271.64	SBI Bank	HDFC Bank	5557.02	Delhi	Female	Transfer	Success	12:14.0	Tablet	Phone Nur	Amazon	Food	21	Scheduled	USD	1.23E+11	9.88E+11		
2	TXN00002	1064.63	ICICI Bank	SBI Bank	9753.32	Bangalore	Male	Payment	Success	15:02.0	Laptop	QR Code	Zomato	Travel	22	Instant	EUR	1.23E+11	9.88E+11		
3	TXN00003	144.15	Axis Bank	Axis Bank	7597.35	Hyderabad	Female	Transfer	Success	29:39.0	Mobile	UPI ID	Swiggy	Bill Payme	23	Scheduled	GBP	1.23E+11	9.88E+11		
4	TXN00004	612.89	HDFC Bank	ICICI Bank	2327.84	Mumbai	Male	Payment	Success	27:36.0	Tablet	Phone Nur	IRCTC	Others	24	Instant	INR	1.23E+11	9.88E+11		
5	TXN00005	743.32	SBI Bank	HDFC Bank	1136.84	Delhi	Female	Transfer	Failed	06:22.0	Laptop	QR Code	Flipkart	Shopping	25	Scheduled	USD	1.23E+11	9.88E+11		
6	TXN00006	665.72	ICICI Bank	SBI Bank	765.1	Bangalore	Male	Payment	Success	06:54.0	Mobile	UPI ID	Amazon	Food	26	Instant	EUR	1.23E+11	9.88E+11		
7	TXN00007	877.72	Axis Bank	Axis Bank	7548.03	Hyderabad	Female	Transfer	Success	36:49.0	Tablet	Phone Nur	Zomato	Travel	27	Scheduled	GBP	1.23E+11	9.88E+11		
8	TXN00008	512.54	HDFC Bank	ICICI Bank	6875.7	Mumbai	Male	Payment	Success	14:17.0	Laptop	QR Code	Swiggy	Bill Payme	28	Instant	INR	1.23E+11	9.88E+11		
9	TXN00009	171.02	SBI Bank	HDFC Bank	5705.81	Delhi	Female	Transfer	Success	02:48.0	Mobile	UPI ID	IRCTC	Others	29	Scheduled	USD	1.23E+11	9.88E+11		
10	TXN00010	288.03	ICICI Bank	SBI Bank	4237.42	Bangalore	Male	Payment	Failed	03:01.0	Tablet	Phone Nur	Flipkart	Shopping	30	Instant	EUR	1.23E+11	9.88E+11		
11	TXN00011	761.47	Axis Bank	Axis Bank	2.1	Hyderabad	Female	Transfer	Success	32:18.0	Laptop	QR Code	Amazon	Food	31	Scheduled	GBP	1.23E+11	9.88E+11		
12	TXN00012	881.5	HDFC Bank	ICICI Bank	7861.46	Mumbai	Male	Payment	Success	14:28.0	Mobile	UPI ID	Zomato	Travel	32	Instant	INR	1.23E+11	9.88E+11		
13	TXN00013	1202.84	SBI Bank	HDFC Bank	5598.1	Delhi	Female	Transfer	Success	51:36.0	Tablet	Phone Nur	Swiggy	Bill Payme	33	Scheduled	USD	1.23E+11	9.88E+11		
14	TXN00014	1959.25	ICICI Bank	SBI Bank	3249.57	Bangalore	Male	Payment	Success	59:53.0	Laptop	QR Code	IRCTC	Others	34	Instant	EUR	1.23E+11	9.88E+11		
15	TXN00015	1057.03	Axis Bank	Axis Bank	2883.52	Hyderabad	Female	Transfer	Failed	19:54.0	Mobile	UPI ID	Flipkart	Shopping	35	Scheduled	GBP	1.23E+11	9.88E+11		
16	TXN00016	445.86	HDFC Bank	ICICI Bank	58.02	Mumbai	Male	Payment	Success	58:44.0	Tablet	Phone Nur	Amazon	Food	36	Instant	INR	1.23E+11	9.88E+11		
17	TXN00017	1169.9	SBI Bank	HDFC Bank	8192.47	Delhi	Female	Transfer	Success	44:51.0	Laptop	QR Code	Zomato	Travel	37	Scheduled	USD	1.23E+11	9.88E+11		
18	TXN00018	468.41	ICICI Bank	SBI Bank	8947.62	Bangalore	Male	Payment	Success	08:57.0	Mobile	UPI ID	Swiggy	Bill Payme	38	Instant	EUR	1.23E+11	9.88E+11		
19	TXN00019	1680.49	Axis Bank	Axis Bank	8755.97	Hyderabad	Female	Transfer	Success	19:38.0	Tablet	Phone Nur	IRCTC	Others	39	Scheduled	GBP	1.23E+11	9.88E+11		
20	TXN00020	9/21/2024	1041.4	HDFC Bank	ICICI Bank	5189.37	Mumbai	Male	Payment	Failed	48:15.0	Laptop	QR Code	Flipkart	Shopping	40	Instant	INR	1.23E+11	9.88E+11	
21	TXN00021	10/22/202	82.31	SBI Bank	HDFC Bank	5917.7	Delhi	Female	Transfer	Success	29:39.0	Mobile	UPI ID	Amazon	Food	41	Scheduled	USD	1.23E+11	9.88E+11	
22	TXN00022	11/23/202	352.02	ICICI Bank	SBI Bank	4431.16	Bangalore	Male	Payment	Success	01:29.0	Tablet	Phone Nur	Zomato	Travel	42	Instant	EUR	1.23E+11	9.88E+11	
23	TXN00023	12/24/202	625.1	Axis Bank	Axis Bank	6280.01	Hyderabad	Female	Transfer	Success	52:19.0	Laptop	QR Code	Swiggy	Bill Payme	43	Scheduled	GBP	1.23E+11	9.88E+11	
24	TXN00024	1/25/2024	789.62	HDFC Bank	ICICI Bank	2415.86	Mumbai	Male	Payment	Success	18:02.0	Mobile	UPI ID	IRCTC	Others	44	Instant	INR	1.23E+11	9.88E+11	
25	TXN00025	2/26/2024	1912.19	SBI Bank	HDFC Bank	1482.52	Delhi	Female	Transfer	Failed	03:35.0	Tablet	Phone Nur	Flipkart	Shopping	45	Scheduled	USD	1.23E+11	9.88E+11	

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UPI+Transactions.xlsx - Sheet1

12345678910111213141516171819202122232425

Dashboard Created:

